Thermodynamic description of the Cu–Ni system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Cu–Ni binary system as calculated with pycalphad from the SGTE dataset CuNi-87Jan.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key.

1. Introduction

The Cu–Ni system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as isomorphous, solid miscibility gap. The thermodynamic description is the SGTE collection entry (Hallstedt, 2025), which reproduces the assessment of A. Jansson, TRITA-MAC 340, KTH, Stockholm 1987. The calculations use pycalphad 0.11.2 at 1 atm.

Pure Cu melts at 1357.8 K. Pure Ni melts at 1728.2 K.

2. Phase diagram

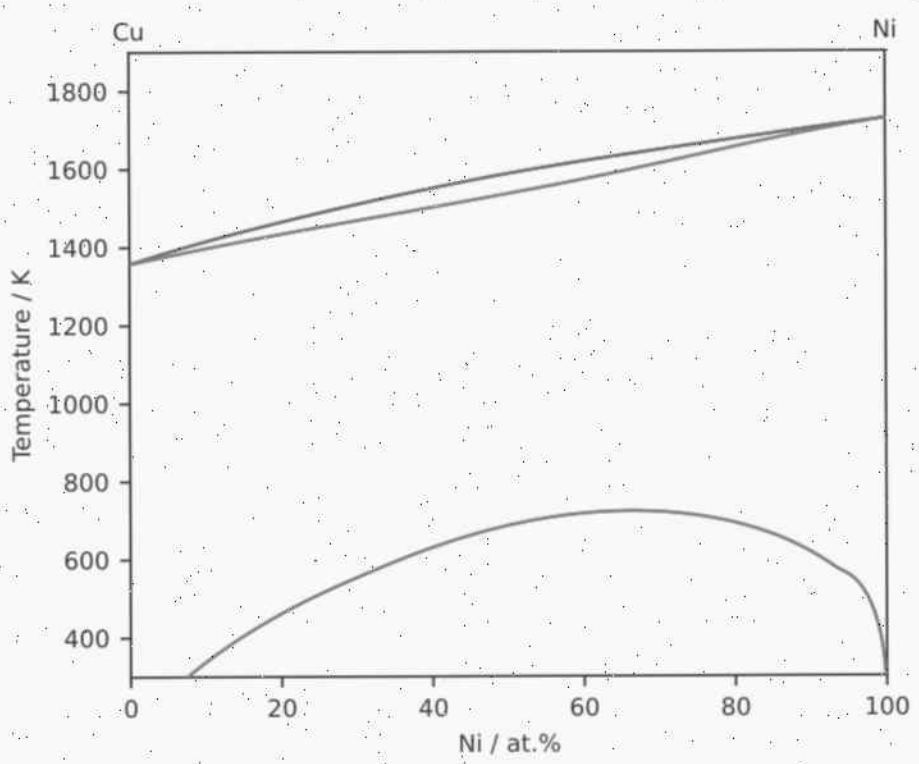


Fig. 1. Calculated Cu–Ni phase diagram.

No invariant reaction occurs between 300 K and the melting points; Cu and Ni form a continuous solid solution at high temperature.

Table 1. Liquidus and solidus temperatures in the Cu–Ni system (calculated).

at.% Ni	T liquidus / K	T solidus / K
20	1464.4	1433.5
40	1550.7	1501.7
60	1619.1	1572.5
80	1675.6	1655.0

3. Thermodynamic properties

At 1800 K the integral mixing enthalpy of the liquid has its maximum, 2.95 kJ/mol, at 54.0 at.% Ni.

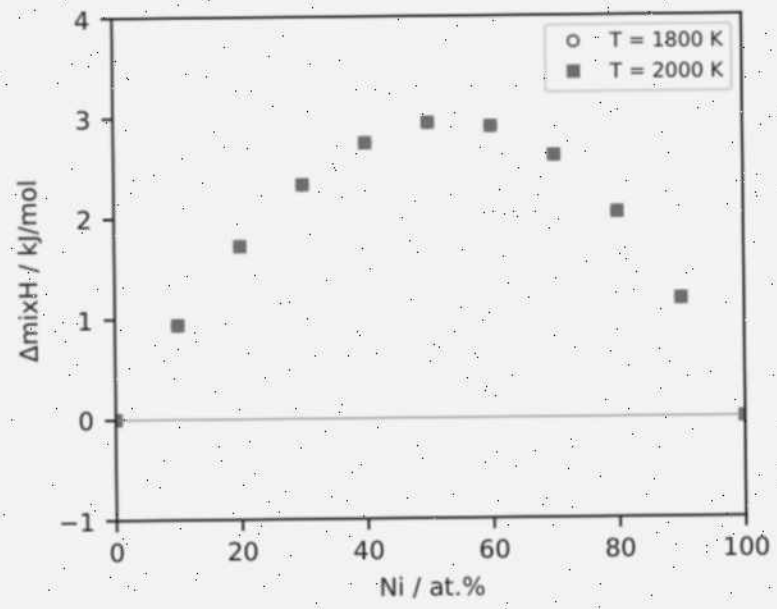


Fig. 2. Calculated mixing enthalpy of the liquid at 1800 K and 2000 K.

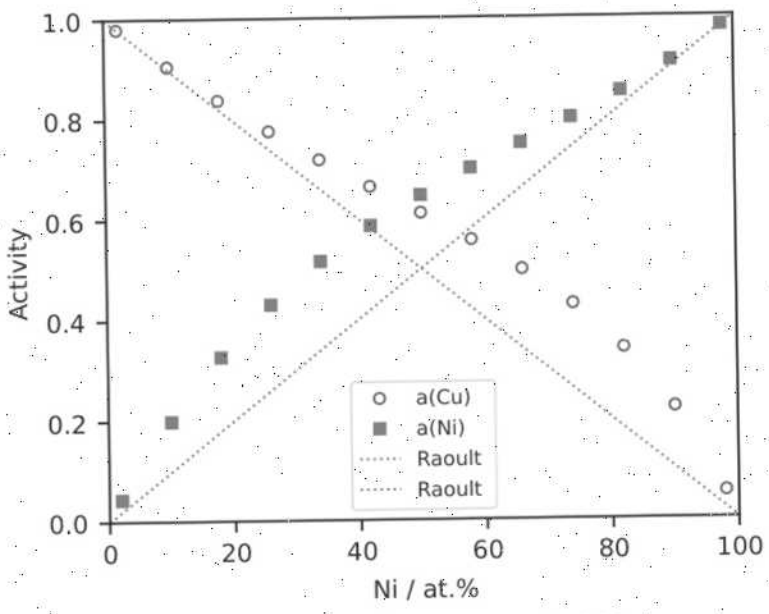


Fig. 3. Calculated activities in the liquid at 1800 K (reference: pure liquids).

Table 2. Redlich–Kister parameters of the liquid (Cu–Ni).

	Liquid: L(ν) = a + b·T + c·T·ln T		
ν	a	b	c
L0	11760.0	1.084	0.0
L1	-1671.8	0.0	0.0

4. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.